Year 12 Maths - Coordinate geometry

Similar questions - Perpendicular bisectors and tangents

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The points A and B have coordinates $(1,2)$ and $(7,p)$ respectively, where p is a positive constant.

(a) Find, in terms of p, the coordinates of the midpoint of AB and the gradient of AB. **[2]**

(b) Show that the perpendicular bisector of AB has equation $2(p-2)y+12x=p^2+44$. **[4]**

(c) The perpendicular bisector of AB passes through the point $(5,0)$. Find the value of p. **[2]**

(d) For this value of p, find the area of the triangle formed by the perpendicular bisector and the two coordinate axes. **[3]**

2 The line $y=mx+3$, where m is a constant, is a tangent to the curve $y=x^2-2x+7$.

(a) Show that $x^2-(m+2)x+4=0$ at any point where the line meets the curve. **[2]**

(b) Hence find the two possible values of m. **[3]**

(c) For the positive value of m, find the coordinates of the point of contact. **[2]**

(d) For the negative value of m, find the x-coordinate of the point where the tangent crosses the x-axis. **[2]**

Question 1

(a) Midpoint $\left(4, \frac{2+p}{2}\right)$. Gradient of $AB = \dfrac{p-2}{6}$

(b) Perpendicular gradient $= -\dfrac{6}{p-2}$. $y - \frac{2+p}{2} = -\dfrac{6}{p-2}(x-4)$. Multiply by $2(p-2)$: $2(p-2)y - (p+2)(p-2) = -12x+48$, so $2(p-2)y+12x = p^2-4+48 = p^2+44$

(c) Substitute $(5,0)$: $60 = p^2+44$, so $p^2 = 16$ and $p = 4$

(d) With $p = 4$: $4y+12x = 60$, i.e. $y = 15-3x$. Intercepts $(5,0)$ and $(0,15)$. Area $= \frac{1}{2} \times 5 \times 15 = 37.5$

Question 2

(a) $x^2-2x+7 = mx+3$, so $x^2-(m+2)x+4 = 0$

(b) Tangent means $b^2-4ac = 0$: $(m+2)^2-16 = 0$, so $m+2 = \pm 4$ and $m = 2$ or $m = -6$

(c) $m = 2$: $x^2-4x+4 = 0$, so $x = 2$ and $y = 2(2)+3 = 7$. Point $(2,7)$

(d) $y = -6x+3$ crosses the x-axis when $x = \frac{1}{2}$